

Mikhael Groschitz

Data Engineer | SQL Server · Azure · Databricks | Python · PySpark · Lakehouse (Delta) · ETL/ELT

Porto Alegre, RS, Brazil · +55 51 99200-0475 · mgroschitz@gmail.com

linkedin.com/in/mikhaelgroschitz · github.com/Mikhael-Groschitz · mgroschitz.dev

SUMMARY

Data engineer focused on SQL Server, Azure, and Databricks. I am currently responsible for the database structure of a production CRM at Grupo Facta — modeling, views, indexes, constraints, and optimization of critical queries — and I maintain Python automations that support day-to-day operations. Outside of work, I build end-to-end pipelines on Azure (Data Factory, ADLS Gen2, Azure SQL) and on Spark with Delta Lake, all public on GitHub. Certified in Databricks Fundamentals.

TECHNICAL SKILLS

Databases and SQL: SQL Server, T-SQL, Azure SQL Database, query optimization (execution plans, indexing), dimensional modeling, SCD Type 2, CDC; PostgreSQL, MongoDB

Cloud and platforms: Azure (ADLS Gen2, Data Factory, Azure SQL), Databricks, Bicep (IaC)

Processing and orchestration: Apache Spark (PySpark), Spark Structured Streaming, Delta Lake, Airflow, Kafka, dbt

Architecture and modeling: Data Warehouse, Lakehouse, medallion architecture (bronze/silver/gold), ETL/ELT data pipelines

Languages: Python (PySpark, Pandas, NumPy), SQL, T-SQL

Quality and practices: Data quality, pytest, Git/GitHub, GitHub Actions (CI), Docker

BI: Power BI

EXPERIENCE

Backend Data Analyst (working in Data Engineering) · Credware Tecnologia (Grupo Facta)

Mar 2026 – Present

Stack: SQL Server, T-SQL, Python, Databricks

- Redesigned the data model of the legal CRM in SQL Server (40+ tables, ~120 views), reducing the runtime of the main routines by ~35%; deployed to production with the DBA team, with no rollback.
- Optimized ~50 critical T-SQL queries (execution plans, indexes, and rewriting): queries went from ~8s to ~1.5s, with ~45% less CPU during peak hours.
- Developed and maintain ~15 Python automations, with monitoring and logging, that replaced manual routines in the legal department (estimated savings of 25 to 30 hours/month).
- Mapped ~10 systems and their data flows, and take part in the CRM's architecture and refactoring decisions, assessing the impact on processes, data, and users.
- Act as the liaison between business, data, and development: translated 30+ requests into technical requirements and documented ~25 business rules.

Junior Data Analyst · Escritório Contábil Fiscal Escofi

Nov 2024 – Mar 2026

Stack: SQL, PostgreSQL, Python (Pandas, NumPy), Power BI

- Optimized ~30 queries and extraction scripts in SQL and Python on datasets of 1 to 5 million rows, reducing data preparation time from ~3h to ~1h.
- Applied anomaly detection to higher-materiality accounts, flagging ~20 atypical entries per cycle for review and reducing audit triage time by ~35%.
- Supported ~5 predictive models for cash flow and profitability (30- to 90-day horizons, evaluated with MAPE) and ~10 Power BI dashboards for ~20 users, reducing management report turnaround from ~2 days to ~4 hours.

IT Support Analyst · PAYER

Oct 2023 – Oct 2024

- Analyzed tickets to identify root causes of recurring issues, with metrics dashboards and insights shared with the development team.

DATA ENGINEERING PROJECTS (GITHUB)

ANP Prices ELT Pipeline — Azure · github.com/Mikhael-Groschitz/anp-fuel-prices-elt-azure

- Metadata-driven Data Factory, ADLS Gen2, Azure SQL, T-SQL, Bicep, and Power BI over 20 years of fuel prices from ANP, Brazil's oil, gas, and biofuels regulator (219 files, 35.3M rows in staging and 32.2M in the fact table) in a dimensional DW. Historical load of ~3h (initial projection: 15h) after UDF inlining, clustered index, and set-based SCD2; 8 quality checks with logging.

SQL Server CDC to Dimensional Data Warehouse · github.com/Mikhael-Groschitz/sqlserver-cdc-to-dw

- Native CDC feeding a DW in T-SQL and Python: incremental extraction by LSN with a watermark in the same transaction, SCD Type 2 with a filtered unique index, soft delete in the fact table to preserve the audit trail, 5 quality checks, and tests in pytest.

CNPJ Lakehouse — Spark, Airflow, and Delta (local) · github.com/Mikhael-Groschitz/cnpj-lakehouse-spark-airflow

- Airflow, PySpark, Delta Lake, MinIO, and DuckDB on Docker over ~30 million establishments from Brazil's Federal Revenue Service (Receita Federal), from bronze to gold in ~2h 40min. Partitioning by state (UF) reduced 5,631 files to 103; 99.99999% of CNPJs validated and 30 scenarios in pytest.

Credit Proposals Streaming — Kafka, Spark, and Delta (local) · github.com/Mikhael-Groschitz/credit-proposals-streaming-mongodb-kafka

- MongoDB change streams → Kafka → Spark Structured Streaming → Delta (bronze/silver/gold), with resume token and Mongo × gold reconciliation. The quality checks detected 508 invalid transitions, reduced to 2 residual cases; a broadcast join removed the scalability bottleneck.

Expectations vs. Actuals — Central Bank of Brazil (Python and dbt) · github.com/Mikhael-Groschitz/bcb-macro-elt-python-dbt

- Idempotent ingestion from the SGS and Focus APIs with watermark and 90-day lookback (~166K expectations), SCD Type 2 snapshot, and a forecast error mart in dbt; 65 Python tests, 69 dbt tests, and CI on GitHub Actions.

Multi-backend Data Observability · github.com/Mikhael-Groschitz/data-observability-multi-backend

- Service that monitors tables from other pipelines (DuckDB and Postgres): freshness, nulls, and schema computed inside the database (1 row transferred even for a 500K-row table), YAML contracts, rolling-median baseline, and webhook alerts; 114 tests.

EDUCATION AND CERTIFICATIONS

Postgraduate Program in Software Architecture, Data Science, and Cybersecurity · PUCPR **Aug 2025 – Oct 2026 (expected)**

Technologist Degree in Systems Analysis and Development · Anhanguera

Jan 2023 – Jun 2025

Databricks Academy Accreditation — Databricks Fundamentals · Databricks

Sep 2026

Languages: Portuguese (native) · English (full professional proficiency) · Spanish (limited working proficiency)